

Does anodal transcranial direct current stimulation over the primary motor cortex really enhance motor sequence learning?

Short/running title

Does tDCS really enhance motor sequence learning?

Authors

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Abstract

Background

Transcranial direct current stimulation (tDCS) has traditionally been assumed to modulate the brain by directly polarizing cortical neurons. However, recent work suggested that the electric field reaching the cortex is weak, whereas the field in the scalp is strong enough to stimulate peripheral nerves, potentially influencing cortical excitability indirectly. In this study, we aimed to replicate earlier findings that anodal tDCS enhances motor sequence learning (MSL) and to investigate whether the effects are primary driven by cortical polarization, peripheral nerve stimulation, or both.

Methods

In a double-blind, pre-registered study including 99 adults, we examined the effect of 1mA anodal tDCS over the primary motor cortex (M1) on MSL using a three-session serial reaction time task (SRTT). Alongside sham, we added an additional control condition to block peripheral nerve input using BL10 gel and thereby isolate peripheral from cortical contributions to tDCS effects.

Results

We observed significant within and across-session improvements in MSL in all conditions ($p < 0.0001$). However, no significant differences were observed between stimulation conditions ($p=0.94$).

Conclusions

Given the lack of a significant effect of tDCS on MSL, we could not address our main research question regarding the underlying neurophysiological mechanisms. Nonetheless, our findings contribute to the evidence that tDCS motor effects are unlikely to be as large as previously suggested and add to the body of literature that sparked discussion regarding the reliability of tDCS as a neuromodulation technique.

Clinical trial registration

The study was registered on ClinicalTrials.gov ([NCT04577677](https://clinicaltrials.gov/ct2/show/study/NCT04577677)) as ‘Understanding Neural Mechanisms Behind tDCS’.

Introduction

Transcranial direct current stimulation (tDCS) is a widely used noninvasive neuromodulation method that involves the application of a low intensity direct current (DC) over scalp electrodes. Early results demonstrating the neuromodulatory potential of the technique(1) sparked a surge of interest among neuroscientists, neurologists, and psychiatrists. Due to its noninvasive nature, low cost, and ease of application compared to other neuromodulation techniques, the field of tDCS has grown rapidly. Despite the large amount of research and numerous studies investigating the clinical applications of the technique, the underlying neurophysiological mechanisms remain poorly understood.

Initially, it was hypothesized that anodal tDCS elicits its neuromodulatory effect by increasing cortical excitability due to depolarization of the resting membrane potential of cortical neurons(1). However, our computational modeling study showed that only a small portion of the applied current effectively reaches the cortex, as most of the current is shunted by highly conductive tissues in between the stimulation electrodes and the targeted cortex such as the scalp, skull, and cerebrospinal fluid(2). We demonstrated that when applying 2 mA tDCS, the resulting electric field strength in the cortex is less than 0.5 V/m. Assuming the polarization length constant of excitatory cortical neurons(3) to be 0.2, a 1 V/m extracellular electric field should be able to cause a membrane potential shift of 0.2 mV at the soma. This means that the resulting electric field in the cortex of less than 0.5 V/m can only induce a depolarization of the resting membrane potential of up to 0.1 mV. Despite ongoing research on this topic, it remains unclear how such a small shift in the resting membrane potential could account for significant changes in cortical excitability as reported in previous tDCS studies(1).

In an opinion piece, we addressed this question and proposed an alternative mechanism of tDCS, known as the transcutaneous tDCS mechanism(4). In our previously mentioned computational study, we also found that in addition to the unexpectedly weak cortical electric field, the resulting electric field in the scalp underneath the stimulation electrodes can reach levels up to 20 V/m(2,5). This is strong enough to elicit action potentials in peripheral nerves that innervate the scalp(6) as evidenced by subjects of tDCS studies often experiencing tingling, burning, or itching sensations during stimulation(7,8). In addition to providing input to the somatosensory cortex, stimulation of these cranial and cervical nerves also activates structures in the brainstem where these nerves enter the brain(9,10) such as the locus coeruleus (LC). As activation of subcortical nodes, such as LC, is known to increase the release of the norepinephrine (NE) and thereby enhance excitability across the cortex, stimulation of peripheral nerves in the scalp during tDCS can also increase cortical excitability, albeit indirectly.

To determine whether the neuromodulatory effects of tDCS are driven by the resulting electric field in the cortex, as originally hypothesized, or by an increase in cortical excitability resulting from stimulation of peripheral nerves in the scalp, it is essential to find a means of isolating and selectively activating the transcranial or the transcutaneous mechanism without engaging the other. We therefore developed a novel tDCS control condition in which the transcutaneous mechanism is blocked by anesthetizing the peripheral nerves in the scalp using the BL10 anesthetic gel(7).

In this study, we aimed to determine the relative contributions of the transcranial and transcutaneous mechanisms by comparing the effects of regular tDCS, in which both mechanisms are active, with this novel control condition in which only the transcranial mechanism is active. If the observed effects of regular tDCS were to diminish in the control condition, this would indicate that the transcutaneous mechanism plays a significant role in mediating tDCS effects. Conversely, if the effects were to be maintained in the control condition, this would suggest that the transcranial mechanism is primarily responsible for the observed effects of tDCS.

Yet, before being able to evaluate the relative contributions of the neurophysiological mechanisms, our first aim was to replicate the effect of tDCS on motor learning(11,12). More specifically, we aimed to replicate the effect of anodal tDCS over the primary motor cortex (M1) on MSL in the serial reaction time task (SRTT) compared to sham stimulation(13–18). Although the effects of tDCS on MSL is considered one of the most consistent and reliable effects of tDCS(11,12) and many studies have demonstrated profound effects(13–18), it is important to note that not all studies observed significant effect(19) and one study even reported that anodal tDCS over M1 impaired performance on the SRTT(20). Furthermore, among the studies showing significant effects, the timing of tDCS-induced improvements varied considerably, with some occurring during practice and concurrent stimulation(16,17), some immediately afterward(15), and some only after a 24-hour consolidation following practice(13).

Despite some studies showing these early effects of tDCS either during or immediately after practice, many studies have highlighted that the effect of tDCS on motor learning is more evident after several days of task practice with concurrent stimulation(11,15). In fact, a meta-analysis on the effect of multi-session anodal tDCS over M1 on SRTT performance indicated that at least three sessions are required to induce a significant effect of tDCS compared to sham (21) when tested in a retest immediately after task practice and stimulation (standardized mean difference: SMD=0.97), or after a 24h-consolidation interval (SMD=1.68), with SMD an estimate for the effect size of tDCS compared to sham(21).

Accordingly, we aimed to replicate the effect of tDCS on motor sequencing learning in an experimental design including three practice sessions with concurrent stimulation including tests immediately after stimulation, as well as after 24h-consolidation. Although some studies suggested that the effects of tDCS may be more pronounced at higher stimulation amplitudes(22), we chose to apply 1 mA to stay aligned with previous findings and ensure a methodologically sound replication. We hypothesized that 1mA anodal tDCS over M1 can significantly improve MSL performance compared to sham stimulation, when measured in the test immediately after stimulation in the final practice session. Based on this effect, we aimed to test our main hypothesis on relative contribution of the underlying neurophysiological mechanisms by comparing the effect of anodal tDCS to the observed effects of our novel control condition in which the transcutaneous mechanism is blocked.

Methods and Materials

In a large-scale, double-blind, randomized, between-subjects, pre-registered study, we investigated the effect of 1 mA anodal tDCS over left M1 on MSL. In total, 99 young healthy adults (M: 37%, F: 63%, age: 23 ± 3 years) were included in the study. Upon inclusion, participants were assigned to one of three stimulation groups: anodal tDCS ($n_{\text{tDCS}}=33$) group, the novel control condition in which the transcutaneous tDCS mechanism was blocked using the BL10 anesthetic gel(7), ($n_{\text{tDCS+BL10}}=33$), or the sham control ($n_{\text{sham}}=33$) group.

Ethical approval

This study was approved by the UZ/KU Leuven Research Ethical Committee (S63709) and performed in accordance with the ethical principles outlined in the Declaration of Helsinki. The study was registered on ClinicalTrials.gov ([NCT04577677](https://clinicaltrials.gov/ct2/show/study/NCT04577677)) and preregistered on [Open Science Framework](https://osf.io/). Prior to preregistration, a pilot study including twelve subjects was performed to optimize the MSL paradigm and the tDCS stimulation parameters.

Subjects

An a-priori power analysis (95% power, $\alpha = 0.05$, two-sided, expected effect size Cohen's $d = 1$) indicated a minimum of 33 subjects per group to detect tDCS effects on MSL compared to sham, resulting in a total sample size of 99 subjects. A more detailed description of this calculation is provided in our OSF preregistration. All subjects met standard inclusion and exclusion criteria for tDCS motor learning studies and were aged 18–40, right-handed, in good general health, and free from neurological, psychiatric, or musculoskeletal conditions, as well as contraindications to brain stimulation, and lidocaine or benzocaine allergies. Individuals with prior SRTT experience, musical training, gaming or typing expertise, metal implants or piercings near stimulation sites, medication use, or potential pregnancy were excluded. All participants provided written informed consent and followed pre-session guidelines, including adequate rest and abstaining from stimulating substances.

Stimulation conditions

In the anodal tDCS condition, 1 mA anodal tDCS was applied over left M1 (C3) using the Signa-gel filled 3D-printable cup electrodes(23) with a diameter of 19 mm and the neuroConn DC-Stimulator Plus (neurocare Group, Munich, Germany) using Ag/AgCl connector cables. The return electrode was located in the contralateral supraorbital position (FP2), as illustrated in Figure 1C. Prior to electrode placement, the scalp was cleaned to ensure optimal conductivity. Finally, a conductivity test was performed to confirm that the system's impedance was below 5 k Ω before stimulation was started with a 1-second ramp to 1 mA.

For the sham condition, the same electrode setup was used, and a brief stimulation was applied to mimic the sensations of tDCS before it was ramped down again after ten seconds, thereby controlling for potential bias induced by subject's expectation related to the effects of tDCS. In the novel control condition, again the same stimulation electrode setup was used. However, to block the transcutaneous tDCS mechanism, an additional layer of the anesthetic BL10 gel (7), containing 5% lidocaine and 5% benzocaine, was applied on the scalp underneath the cup electrodes precisely matching the electrode's location and surface area to keep the stimulation surface area consistent.

Motor sequence learning paradigm

The MSL paradigm comprised an introduction session followed by three MSL sessions conducted on three consecutive days (Figure 1A). During the introduction session, participants received detailed study information, provided written informed consent, and underwent 10 minutes of tDCS to familiarize themselves with the stimulation prior to the practice sessions.

Serial reaction time tasks

In each MSL session, participants performed a sequenced serial reaction time task(24) (SRTT) while receiving concurrent stimulation (Figure 1C). Each SRTT consisted of sixteen sequenced blocks (Se) as indicated in Figure 1A: $Se_{1,1}$, $Se_{2,2}$, and $Se_{3,2}$, with each block comprising 8 repetitions of an 8-element stimuli (Figure 1B), with 10-second rest intervals between each block to reduce fatigue (Figure 1B). At the start of each session, subjects were instructed on how to perform the SRTT. Four empty squares were displayed on a screen, each corresponding to a key on the keyboard, and a finger of the right hand (Figure 1D). Visual stimuli were presented by filling one the squares (Figure 1D) and subjects were asked to press the correct key with the corresponding finger as quickly and accurately as possible. After each key press, whether correct or incorrect, the next stimulus was immediately presented. Each subject was assigned to practice an 8-element sequences (Suppl. Materials).

SRTT familiarization and baseline performance

At the beginning of the first session, an additional familiarization phase was implemented, consisting of 16 random blocks (Ra) of 64 stimuli each (Figure 1A: Ra_{1,0}), in which the squares were filled in a pseudo-randomized order. These blocks allowed familiarization and controlled for non-sequence-specific visuomotor habituation, with no stimulation applied. In addition, four random blocks were included at the beginning of each session (Figure 1A: Ra_{1,1}, Ra_{2,2}, Ra_{3,2} to familiarize subjects before the sequenced SRTT and thereby minimize initial performance effects. The four additional random blocks performed after task familiarization (Figure 1A: Ra_{1,0}) and before the start of the first sequenced SRTT in session 1, were added to measure baseline performance (Figure 1A: Ra_{1,1}).

MSL improvements across practice sessions

Immediately after each sequenced SRTT (Figure 1A: Se_{1,1}, Se_{2,2}, and Se_{3,2}), potential improvements in MSL performance were assessed in a test by comparing the subject's performance in four sequenced blocks (Figure 1A: Se_{1,2}, Se_{2,3}, and Se_{3,3}) to their performance in four random blocks (Figure 1A: Ra_{1,2}, Ra_{2,3}, and Ra_{3,3}), during which no stimulation was applied.

Data analysis

The MSL paradigm was designed to facilitate MSL over time. Improvements were assessed based on changes in the mean reaction times (RTs), i.e. the time between stimulus presentation and key press response in milliseconds (ms), per block. To avoid bias from incorrect responses, only RTs from correct responses were included when averaging per block. Improvements in task performance across sessions were assessed based on the mean RTs of the four sequenced blocks of the tests at the end of each session (Se_{1,2}, Se_{2,3}, and Se_{3,3}) relative to previous sessions and baseline. Within-session improvements were calculated by subtracting the mean RTs of the sequenced blocks from the mean RTs of the random blocks of the respective tests. As described in the preregistration, subjects showing no improvement in any practice session or with overall key accuracy lower than 85% were considered noncompliant or disengaged and were excluded from further data analysis. Accordingly, a total of eight subjects were excluded (5 in the tDCS group, 3 in the sham group). All analyses were conducted in MATLAB 2019 (MathWorks® R2019b).

Statistical analysis

MSL improvements within and across sessions were compared using unpaired two-sample t-tests with Bonferroni-adjusted significance levels ($\alpha = 0.05/3$). The effect of stimulation condition on MSL immediately after stimulation across sessions was analyzed using a linear mixed model (LMM): MSL immediately after stimulation $\sim 1 + \text{session} + \text{stimulation condition} + \text{session} \times \text{stimulation condition} + (1 | \text{subject})$, with fixed effects for session and stimulation condition, and a random intercept for subject. In addition to the analyses specified in the preregistration, a Bayes factor was calculated to assess the likelihood of the null hypothesis that tDCS does not enhance MSL after three practice sessions compared to sham with respect to the alternative hypothesis that tDCS enhanced MSL. Furthermore, within-session differences between stimulation conditions were analyzed using unpaired two-sample t-tests with Bonferroni correction for multiple comparisons. Next, for each session respectively, the effect of stimulation conditions on online MSL during concurrent stimulation was assessed by comparing block-wise mean reaction times (RTs) across conditions. Mean RTs were averaged across subjects within each condition and baseline-corrected using group-specific baselines. Effects were analyzed using repeated-measures ANOVA with block as a within-subject factor, followed by pairwise comparisons between conditions for each block using unpaired two-sample t-tests. Finally, the effect of 24-hour overnight consolidation on MSL retention was evaluated by comparing performance at the end of each session with performance 24 hours later at the beginning of the subsequent session (Figure 1A: Se_{2,1} and Ra_{2,1}, Se_{3,1} and Ra_{3,1}). Further details are provided in the Supplemental Materials.

Results

Our results indicated that all stimulation groups showed sustained improvements in MSL across all sessions, as well as significant online learning during practice with concurrent stimulation in each session. However, no significant differences were observed between stimulation groups, either after a single session or multiple sessions, during practice or after consolidation.

Baseline performance

After task familiarization, baseline performance was comparable across stimulation groups (mean RT: tDCS=406±33 ms, tDCS+BL10=397±38 ms, and sham=398±35 ms) with no significant differences between conditions: $\Delta_{\text{tDCS-Sham}}=8\text{ms}$, $p=0.474$, $t(56)=0.72$, 95% CI [-11.19, 23.75], Cohen's $d=0.1894$, $\Delta_{\text{tDCS-tDCS+BL10}}=9\text{ms}$, $p=0.342$, $t(59)=0.957$, 95% CI [-9.59, 27.17], Cohen's $d=0.248$, and $\Delta_{\text{tDCS+BL10-sham}}=1\text{ms}$, $p=0.785$, $t(61)=-0.274$, 95% CI [-20.84, 15.83], Cohen's $d=-0.069$.

Continued MSL across sessions

Subjects exhibited continued MSL across sessions (Figure 2), as reflected by significant reductions in mean RT relative to the preceding session for all stimulation conditions (Table 1), except for the tDCS ($p=0.026$) and tDCS+BL10 ($p=0.023$) groups in session 3 which did not reach significance relative to session 2 after multiple-comparison corrections. Improvements over time should not be interpreted as a standalone result but primarily reflect adherence to the task. It is, however, important to show that all groups demonstrated progressive learning with practice each session (Figure 3A). Figure 3B illustrates individual subject performance, with each dot below the diagonal indicating improvement in performance relative to the previous session or baseline. Only a few dots above the diagonal indicated that a small number of subjects did not improve in every session. This highlights an important result for further discussion in this paper: rather than plateauing early on, all groups continued to show improvements throughout the paradigm.

Effect of stimulation on MSL across and within sessions

The LMM fitted on within-session improvements each session (Figure 4), showed continued learning over time, as reflected by the significant main effect of session ($\beta = 39\text{ms}$, $\text{SE} = 3.85$, $t(267) = 10.10$, $p < 0.0001$, 95% CI [31, 46]). However, no significant main effects of stimulation condition were found, with both the tDCS+BL10 ($\beta = -3.82$, $\text{SE} = 17.22$, $t(267) = -0.22$, $p = .825$) and sham group ($\beta = -9.44$, $\text{SE} = 17.61$, $t(267) = -0.54$, $p = 0.592$) not significantly different from the tDCS group. Additionally, within each session, unpaired two-sample t-tests revealed no significant differences in MSL improvements between stimulation conditions (Table 2). These findings were further supported by the Bayes factor ($\text{Bayes}_{\text{tDCS-sham}}=0.153$) being well below 1, providing substantial evidence against the hypothesis that tDCS can enhance MSL, and supporting the conclusion that tDCS had little to no effect compared to sham after three practice sessions.

Effect of stimulation on online MSL

In each session, all stimulation groups showed significant improvements in online MSL, as indicated by significant effects of block in repeated-measures ANOVAs (session 1: $F(15,1305)=6.81$, $p<0.001$; session 2: $F(15,1305)=6.32$, $p<0.001$; session 3: $F(15,1305)=7.67$, $p<0.001$) during concurrent stimulation (Figure 5). However, no interaction effect between block and stimulation condition was observed in any session (session 1: $F(30,1305)=1.04$, $p=0.406$; session 2: $F(30,1305)=1.37$, $p=0.087$; session 3: $F(30,1305)=1.04$, $p=0.408$), indicating no effect of stimulation condition on online learning curves.

Discussion

In this study we aimed to replicate the effect of 1mA anodal tDCS over M1 on MSL in a multiple session serial reaction time task paradigm to determine whether tDCS's neuromodulatory effects are driven by the resulting electric field in the cortex, as originally hypothesized (1), or by an increase in cortical excitability resulting from stimulation of peripheral nerves in the scalp mechanism (4). Despite some earlier concerns regarding replicability(19,20), uncertainty about the effect size (25,26) and the exact timing of tDCS effects on MSL, whether they occur during practice and concurrent stimulation(16,17), immediately afterwards(15), after 24h consolidation(13), or after multiple sessions(11,12), we hypothesized that a significant effect would be evident after three sessions(21).

Contrary to our expectations, we found no evidence that 1 mA anodal tDCS improves MSL, not during practice and concurrent stimulation, nor immediately following the first practice session, after three sessions, or after consolidation (Suppl. Materials). Hence, we were unable to investigate our main hypothesis regarding the underlying neurophysiological mechanisms and determine the relative contributions of the transcranial and the transcutaneous tDCS mechanisms in mediating behavioral effects. Although 1mA tDCS has previously been shown to improve MSL performance (11,12,21), some might argue that the stimulation intensity applied here was insufficient to elicit an effect (27). Yet several other studies have demonstrated that increasing the stimulation amplitude does not necessarily increase tDCS's efficacy (28), and may even reverse its effects(29). While we did not observe a significant effect of tDCS on MSL compared to sham, all stimulation groups exhibited sustained improvements both within and across sessions, indicating that participants had not yet reached maximal performance and continued to improve throughout the paradigm. This suggests that our paradigm was robust and sufficiently sensitive to allow for potential tDCS-related enhancements each session.

A notable distinction from previous studies, however, is the use of the newly developed tDCS cup electrodes(23) rather than conventional rubber patch electrodes. These electrodes were designed to enhance precision, reproducibility, and reliability of tDCS in humans by addressing commonly reported limitations of conventional electrode and provide lower and more stable impedances at the electrode–scalp interface(23) with a resistivity of approximately $\rho=0.3\Omega\text{m}$ (7). Consequently, according to Ohm's law, a lower voltage is required to sustain the same DC current, which may have influenced the observed tDCS outcomes more than originally anticipated. What if tDCS effects depended more on the resulting voltage and the variability of that voltage than on the applied direct current? Although it seems plausible that the voltage has a more proportional impact on the resulting electric field than the current, this alternative point of view seems to have been overlooked by the tDCS field. If this theory holds, the use of the new gel-filled cup electrodes, originally designed to lower impedance at the electrode–scalp interface and improve tDCS efficacy, may in fact have obscured tDCS effects. While this could be an explanation why some studies have failed to replicate earlier findings of studies that used conventional tDCS electrodes (27,30). In addition, it might even explain why some studies using higher stimulation amplitudes did not necessarily induce more profound effects (28). Although this theory could explain some of the inconsistencies in observed tDCS results, it remains speculative and warrants further investigation in future studies.

In conclusion, we aimed to investigate the neurophysiological mechanisms underlying tDCS (4), but were unable to test our hypothesis because we could not replicate earlier findings showing significant effects of tDCS on MSL (13–18). Although MSL was considered one of the most reliable and consistent behavioral outcomes of tDCS at the time this study was designed (12), reviews (26) and recent meta-analyses (25) suggested that the effects of tDCS on motor learning may be smaller and more variable than earlier proposed. The lack of significant effects observed here, despite a substantially larger sample size than the original studies, together with several other studies reporting similar findings (18–20,31–33), further underscores ongoing challenges related to reproducibility in the tDCS field (34) and contributes to the growing literature suggesting that, if tDCS effects on motor learning exist, they are likely small. While variability in behavioral outcomes may partly reflect differences in study paradigms, stimulation parameters, or application methods, the substantial inconsistency across findings highlights the lack of a fundamental understanding of the neurophysiological mechanisms underlying tDCS. For future research, we therefore recommend systematically investigating these mechanisms using animal models (35) designed to distinguish between transcutaneous and transcranial mechanisms underlying tDCS effects. The resulting fundamental insights could subsequently be validated in human clinical studies to further clarify whether, and under which conditions, tDCS induces reliable behavioral effects, thereby informing its potential use in clinical neuromodulation.

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Author Contributions

The authors' contributions were as follows. Silke Kerstens: hypothesis, methodology, experimental design, paradigm optimization, design and programming of scripts for data acquisition, data acquisition, study administration, data analysis and manuscript writing. Luuk van Boekholdt: methodology, experimental design, data acquisition. Hans Vanderheyden and Louise De Smedt: data acquisition. Nina Seminck and Tine Van Bogaert: support in pilot data collection. Genevieve Albouy and Bradley Ross King: methodology, experimental design, scripts for data acquisition, manuscript review. Jean-Jacques Orban de Xivry: methodology and experimental design. Myles Mc Laughlin: hypothesis, methodology and experimental design.

Conflict of Interest

All authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Table 1

Table 1 Two-paired t-tests comparing SRTT performance across sessions per stimulation condition

tDCS	ΔRT [ms]	95% CI [ms]	tstat	p-value	Cohen's d
Baseline – session 1	90	[60, 121]	$t(54) = 5.98$	< 0.0001	1.60
Baseline – session 2	171	[138, 205]	$t(54) = 10.16$	< 0.0001	2.71
Baseline – session 3	218	[190, 246]	$t(54) = 15.50$	< 0.0001	4.14
Session 1 – session 2	81	[39, 123]	$t(54) = 3.88$	0.0003	1.04
Session 1 – session 3	128	[90, 165]	$t(54) = 6.81$	< 0.0001	1.82
Session 2 – session 3	46	[6, 87]	$t(54) = 2.30$	0.026	0.61
tDCS+BL10					
Baseline – session 1	91	[60, 122]	$t(64) = 5.94$	< 0.0001	1.46
Baseline – session 2	177	[147, 206]	$t(64) = 11.92$	< 0.0001	2.93
Baseline – session 3	216	[191, 242]	$t(64) = 17.05$	< 0.0001	4.20
Session 1 – session 2	86	[47, 124]	$t(64) = 4.48$	< 0.0001	1.10
Session 1 – session 3	125	[90, 160]	$t(64) = 7.15$	< 0.0001	1.76
Session 2 – session 3	40	[6, 74]	$t(64) = 2.32$	0.023	0.57
Sham					
Baseline – session 1	84	[56, 112]	$t(58) = 5.99$	< 0.0001	1.55
Baseline – session 2	169	[139, 198]	$t(58) = 11.54$	< 0.0001	2.98
Baseline – session 3	216	[193, 239]	$t(58) = 18.56$	< 0.0001	4.79
Session 1 – session 2	85	[48, 121]	$t(58) = 4.64$	< 0.0001	1.20
Session 1 – session 3	132	[100, 164]	$t(58) = 8.26$	< 0.0001	2.13
Session 2 – session 3	47	[14, 80]	$t(58) = 2.87$	0.0058	0.74

Table 2

Table 2 Two-paired t-tests comparing within-session MSL improvements between stimulation conditions

Session 1	Δ [ms]	95% CI [ms]	tstat	p-value	Cohen's d
tDCS – sham	5.45	[-24, 35]	$t(56) = 0.372$	0.711	0.098
BL10 – sham	1.37	[-29, 32]	$t(61) = 0.090$	0.929	0.023
tDCS – tDCS+BL10	-4.08	[-37, 28]	$t(59) = -0.252$	0.802	-0.065
Session 2					
tDCS – sham	-3.76	[-38, 31]	$t(56) = -0.220$	0.827	-0.058
BL10 – sham	-4.43	[-35, 27]	$t(61) = -0.261$	0.795	-0.066
tDCS – tDCS+BL10	-0.29	[-34, 34]	$t(59) = -0.017$	0.987	-0.004
Session 3					
tDCS – sham	-5.14	[-33, 23]	$t(56) = -0.364$	0.717	-0.096
BL10 – sham	-7.71	[-32, 17]	$t(61) = -0.633$	0.529	-0.160
tDCS – tDCS+BL10	-2.57	[-29, 24]	$t(59) = -0.193$	0.847	-0.050

Figure 1

Figure 1 Study protocol. A) The study protocol consisted of an introduction session and three motor sequence learning (MSL) sessions, during which participants performed a serial reaction time task (SRTT) comprising sequenced (Se) and random (Ra) blocks with concurrent stimulation. After the SRTT, subjects were given a brief two-minute break (B), during which they were asked to indicate their stimulation sensation (Suppl. Materials) on a visual analogue scale (V). Next, participants completed an SRTT test to assess motor sequence learning (MSL) progress both within and across sessions. At the beginning and end of each session, participants completed a psychomotor vigilance task (PVT) without concurrent stimulation to assess a potential effect of stimulation of task practice on vigilance. At the beginning of Sessions 2 and 3, an additional test was added to allow the assessment of 24-hour consolidation effects on motor sequence learning (MSL) retention. Finally, at the end of each session, subjects reported on potential stimulation side effects (Suppl. Materials) in a short questionnaire (Q). At the beginning of session 2 and session 3, an additional test was added to assess 24h-consolidation effects on MSL retention. During SRTT practice and tests, both speed (sp) and accuracy (acc) were measured to assess performance. During the PVT only reaction times in terms of speed (sp) were recorded. B) Each practice SRTT consisted of 16 blocks, with each block comprising 64 key presses. In the sequenced (Se) blocks, this corresponded to 8 repetitions of an 8-element SRTT sequence, whereas in the random (Ra) blocks, it consisted of 64 pseudo-randomized key presses. Between blocks, subjects had a brief 10-second rest period. C) Stimulation was applied concurrently during SRTT practice using Signa gel-filled cup electrodes, with the anode positioned over the left primary motor cortex (M1) and the return electrode placed in the contralateral supraorbital (SO) region. D) During the SRTT, four empty squares were displayed horizontally on a screen, corresponding to four keyboard keys and the four fingers of the right hand (excluding the thumb). During the task, one square was filled to indicate the target, and participants were instructed to respond as quickly and accurately as possible by pressing the corresponding key with the matching finger. Following each key press, whether correct or incorrect, the next stimulus was presented. Rest periods between blocks were indicated by the squares turning red. Two seconds before the start of the next block, the squares turned green, allowing participants to refocus on the task.

Figure 2

Figure 2 Motor sequence learning paradigm results. Mean reaction times (RTs) per block are shown in chronological order, beginning with the familiarization phase, followed by the first sequenced SRTT and subsequent test at the end of session 1. Sessions 2 and 3 each began with a retest, continued with a sequenced SRTT, and ended each with a final test. Data are displayed for tDCS (blue), tDCS+BL10 (red), and sham stimulation (black). Shaded gradients represent 95% confidence intervals, calculated as the standard deviation divided by the square root of the number of subjects per stimulation group.

Figure 3

Figure 3 Motor sequence learning improvements across sessions. (A) Box plots show significant reductions in reaction time at the end of each session compared to baseline and the preceding session across all stimulation conditions. In each box plot, the red line marks the median, the box edges indicate the 25th and 75th percentiles, and whiskers extend to the most extreme values not considered outliers. Dots represent individual participants: tDCS (n = 28, blue), tDCS+BL10 (n = 33, red), and sham (n = 30, black). (B) Scatter plots show mean reaction times for individual participants in the sequenced blocks in test 1 versus baseline (top row), test 2 versus test 1 (middle row), and test 3 versus test 2 (bottom row) to evaluate session-to-session improvements. Dots below the diagonal (black line) indicate faster reaction times (improved performance) relative to the previous test or baseline, consistent with motor sequence learning, while dots above the line indicate slower reaction times.

Figure 4

Figure 4 Effect of stimulation conditions on motor sequence learning measured immediately after stimulation. Motor sequence learning (MSL) improvements were calculated as the difference between the mean reaction times of four sequenced blocks and four random blocks in the test at the end of each session. The bar plots represent group means and show comparable learning levels across the three stimulation conditions: tDCS (n = 28, blue), tDCS+BL10 (n = 33, red), and sham (n = 30, black), with the error bars indicate standard deviations.

Figure 5

Figure 5 Effect of stimulation conditions on online motor sequence learning. Baseline-corrected online motor sequence learning (MSL) curves in (A) session 1, (B) session 2, and (C) session 3 for each stimulation condition: tDCS (ntDCS = 28, blue), tDCS+BL10 (ntDCS+BL10 = 33, red), and sham (nsham = 30, black), including the baseline-corrected four random blocks preceding each sequenced SRTT. Stimulation was applied only during practice on the sequenced SRTTs, with onset indicated by the vertical black lines. Shaded areas represent 95% confidence intervals ($0.95 \times SD / \sqrt{n}$, with n number of subjects per stimulation condition) around the mean RTs for each stimulation condition. Across all sessions, learning curves were very similar between stimulation conditions.